

BEVFusion

Multi-task multi-sensor fusion with unified bird's-eye view representation
IEEE international conference on robotics and automation (ICRA). 2023.

허정원 최승준



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What is 3D Object Detection?

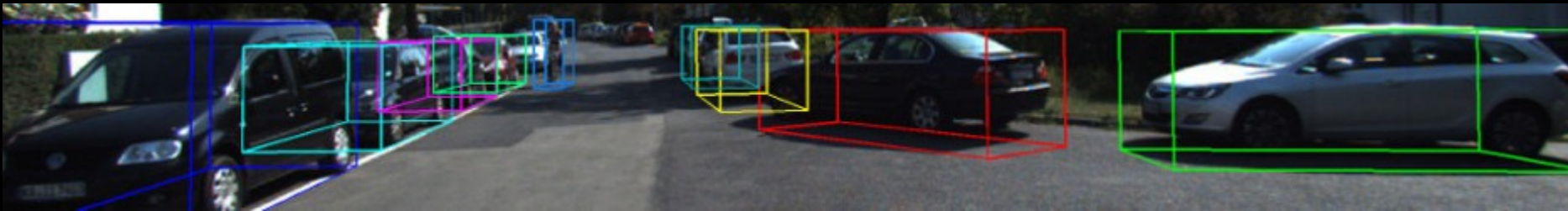
Problem definition

$$\mathcal{B} = f_{det}(\mathcal{I}_{sensor}),$$

$\mathcal{B} = \{B_1, \dots, B_N\}$ is a set of N 3D **object** in a scene,

f_{det} is a 3D object detection **model**,

$\mathcal{I}_{sensor}$ is one or more sensory **inputs**.



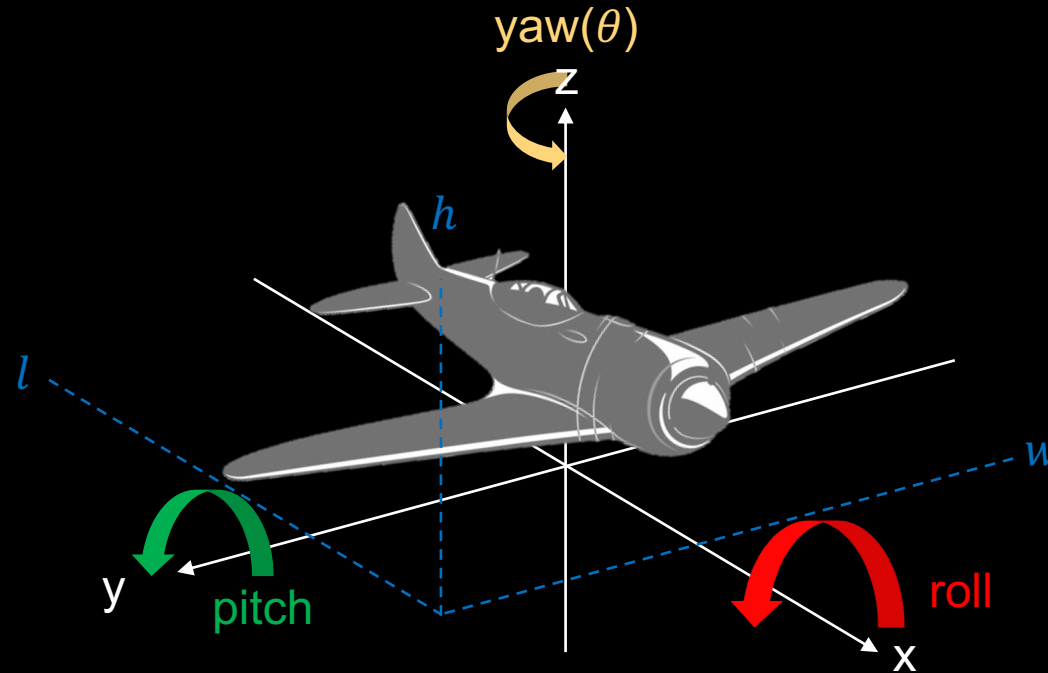
KITTI

[1] Mao, Jiageng, et al. "3D object detection for autonomous driving: A comprehensive survey." *International Journal of Computer Vision* (2023): 1-55.

3D Cuboid

$$B = [x_c, y_c, z_c, l, w, h, \theta, class]$$

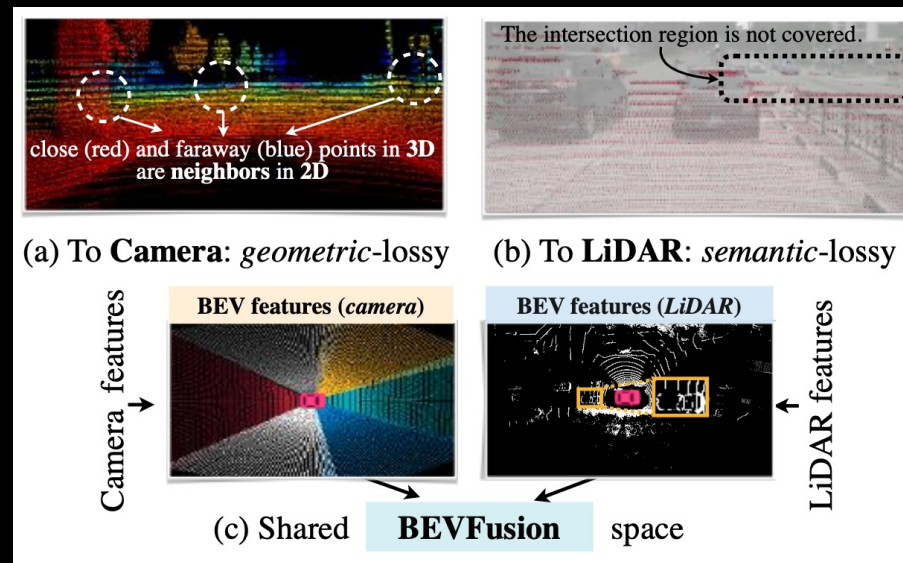
v_x, v_y - velocity



Introduction - Unified Representation

Different sensors provide complementary signals: e.g., cameras capture rich semantic information, LiDARs provide accurate spatial information.

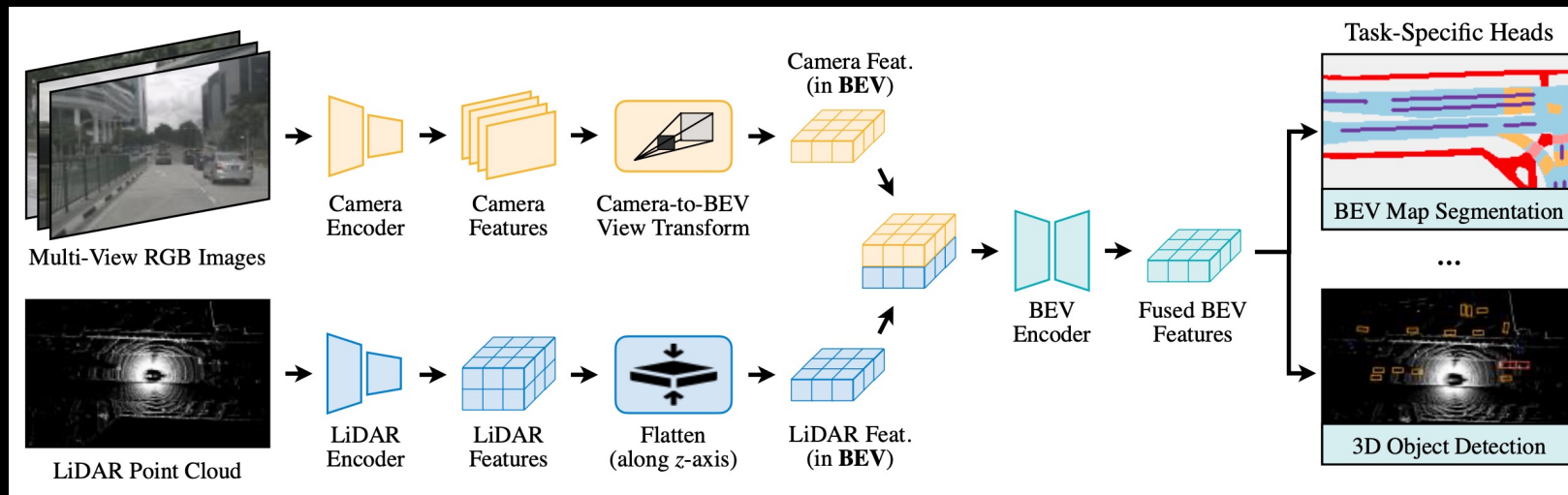
Propose BEVFusion to unify multi-modal features in a shared bird's-eye view (BEV) representation



Introduction - Efficient Camera-to-BEV Transformation

The BEV pooling operation alone takes more than 80% of the model's runtime.

Propose a specialized kernel with precomputation and interval reduction to eliminate this bottleneck.



Methods - Unified Representation

Camera features are in the perspective view, while LiDAR features are typically in the 3D/bird's-eye view.

BEV is friendly to almost all perception tasks since the output space is also in BEV.

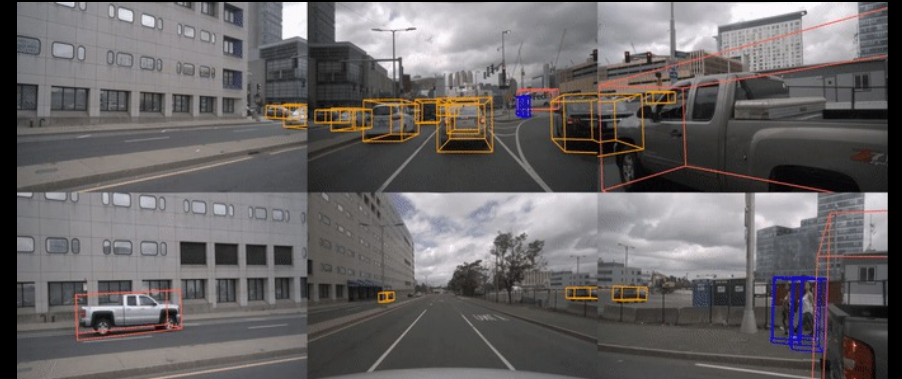
The transformation to BEV keeps both geometric structure (from LiDAR features) and semantic density (from camera features).



Bird's Eye View(LiDAR)



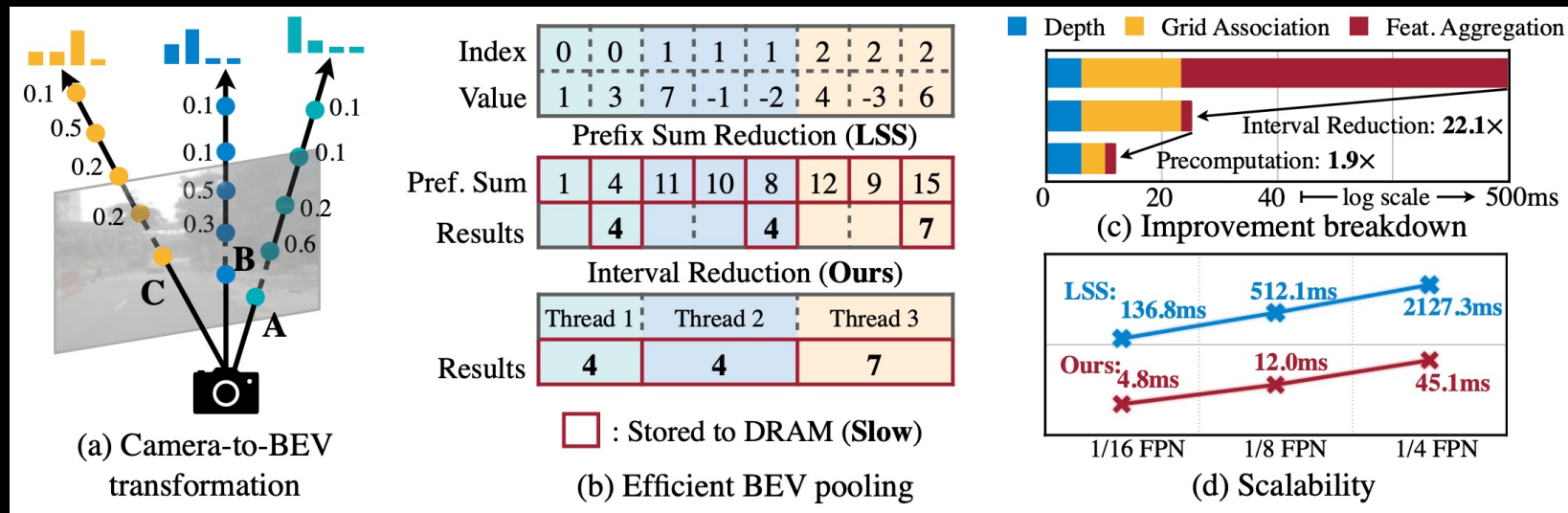
Point View



Cylindrical View

Methods - Efficient Camera-to-BEV Transformation

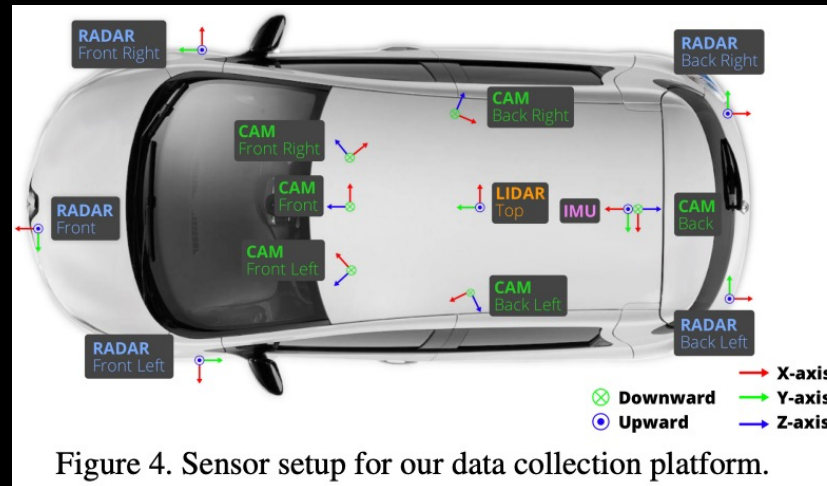
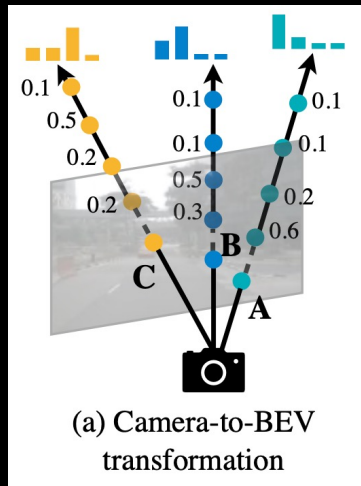
Following LSS, explicitly predict the discrete depth distribution of each pixel.
BEV pooling is surprisingly inefficient and slow.



Methods - Precomputation (BEV Transformation)

The first step of BEV pooling is to associate each point in the camera feature point cloud with a BEV grid. The coordinates of the camera feature point cloud are fixed as long as the camera intrinsics and extrinsics stay the same.

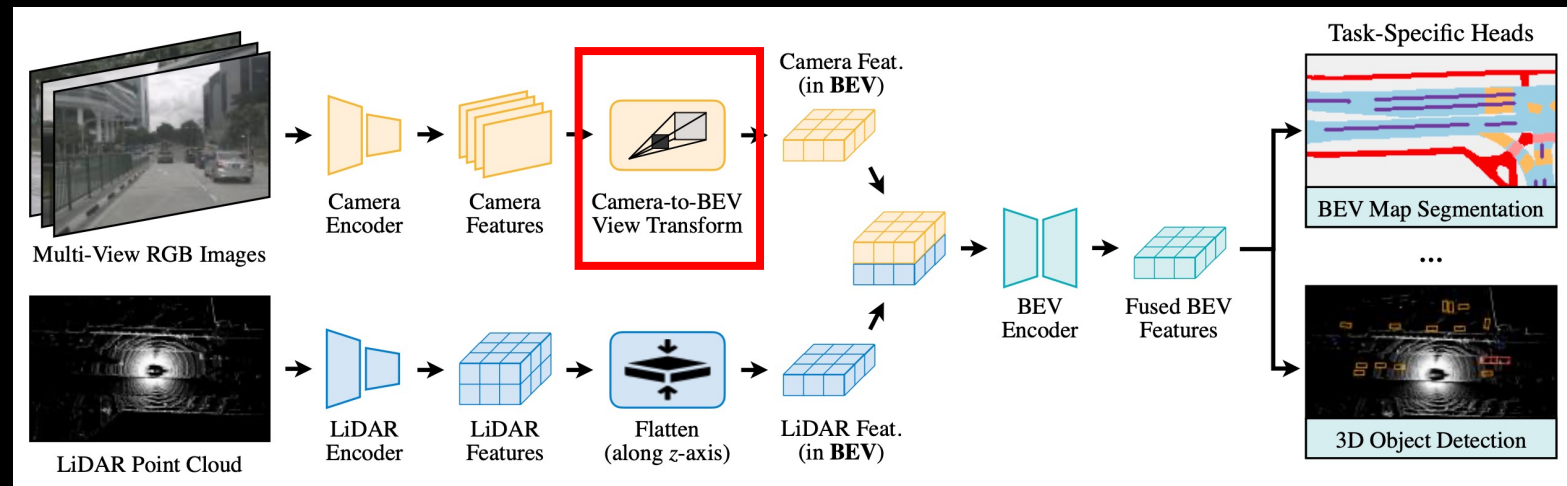
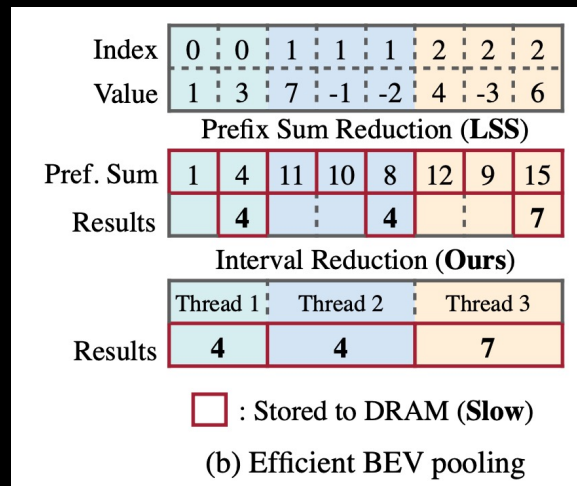
1. Precompute the 3D coordinate and the BEV grid index of each point.
2. Sort all points according to grid indices and record the rank of each point.



Methods - Interval Reduction (BEV Transformation)

The next step of BEV pooling is to aggregate the features within each BEV grid by some symmetric function (e.g., mean, max, and sum).

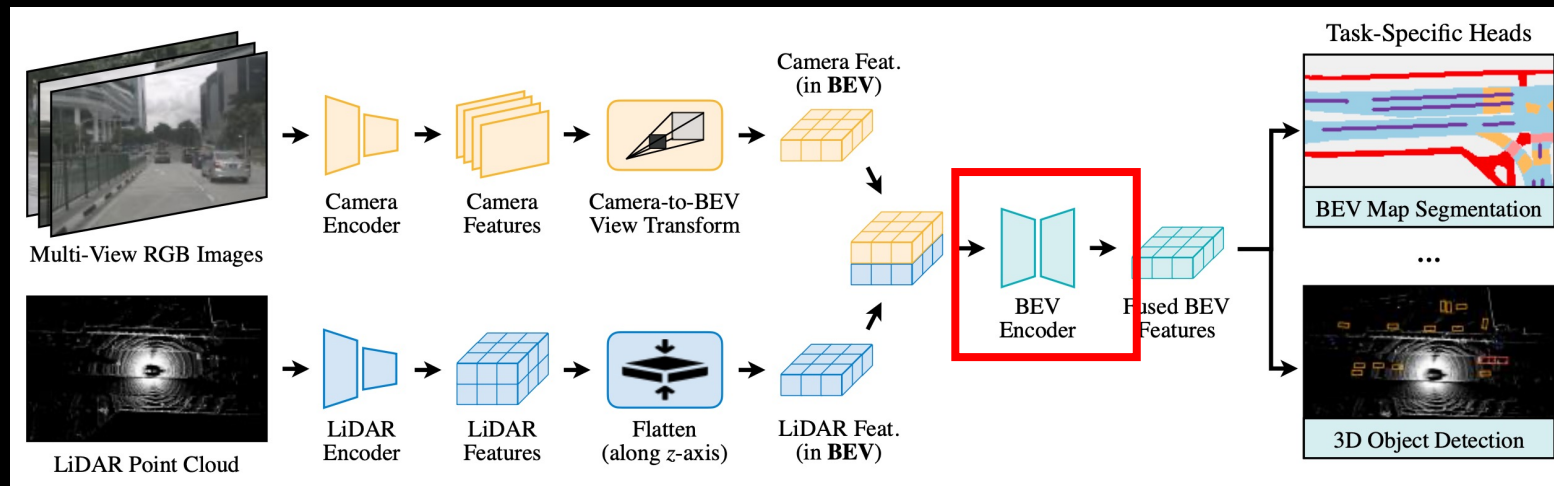
Assign a GPU thread to each grid that calculates its interval sum and writes the result back.



Methods - Fully-Convolutional Fusion

Though in the same space, LiDAR BEV features and camera BEV features can still be spatially misaligned to some extent due to the inaccurate depth in the view transformer.

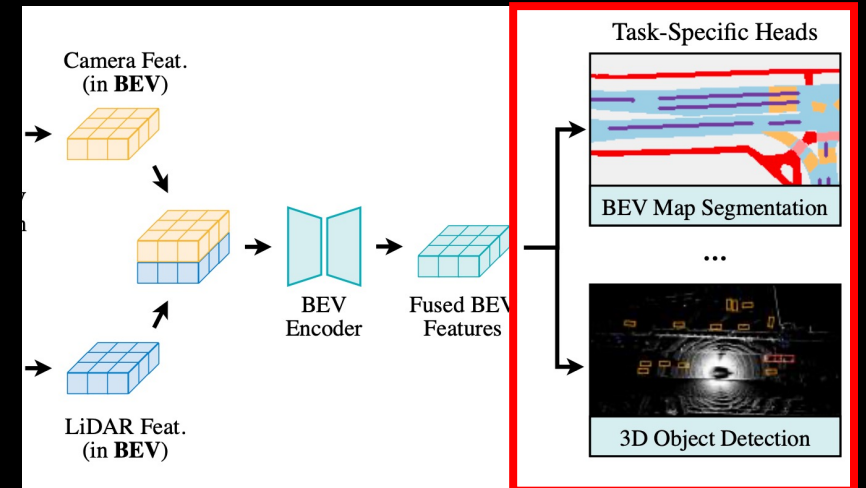
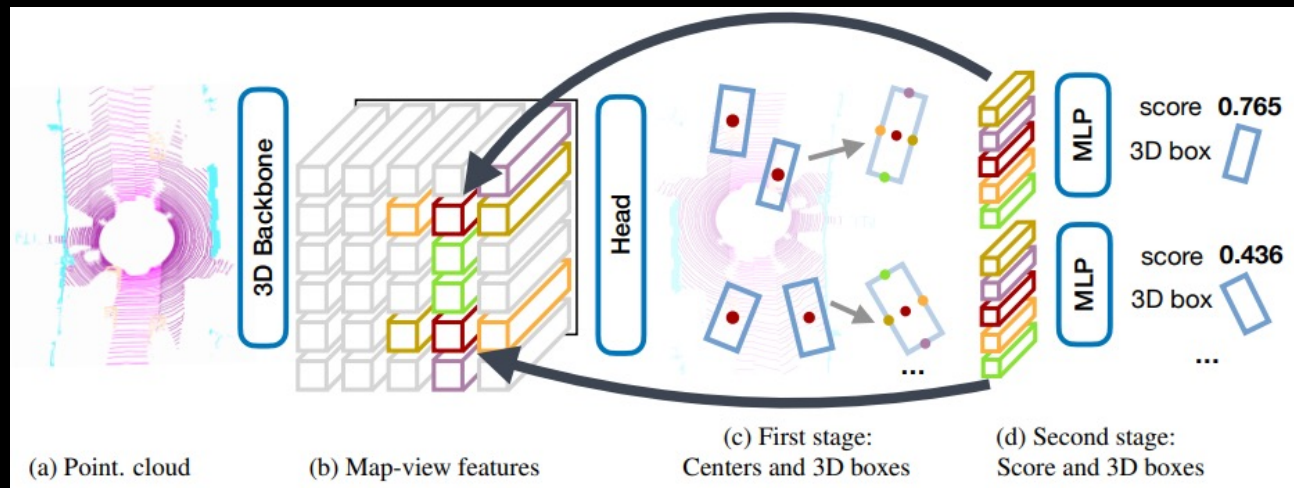
To this end, we apply a convolution-based BEV encoder (with a few residual blocks) to compensate for such local misalignments.



Methods - Multi-Task Heads

A class-specific center heatmap head to predict the center location of all objects and a few regression heads to estimate the object size, rotation, and velocity.

The center-head's goal is to produce a heatmap peak at the center location of any detected object. This head produces a k -channel heatmap $\hat{y}$, one channel for each of k classes.



Experiments - 3D Object Detection

BEVFusion achieves state-of-the-art results on the nuScenes detection benchmark, with close-to-real-time (8.4 FPS) inference speed on a desktop GPU.

BEVFusion also achieves state-of-the-art performance on the Waymo open dataset.

	Modality	mAP (<i>test</i>)	NDS (<i>test</i>)	mAP (<i>val</i>)	NDS (<i>val</i>)	MACs (G)	Latency (ms)
M ² BEV [41]	C	42.9	47.4	41.7	47.0	–	–
BEVFormer [43]	C	44.5	53.5	41.6	51.7	–	–
PointPillars [10]	L	–	–	52.3	61.3	65.5	34.4
SECOND [11]	L	52.8	63.3	52.6	63.0	85.0	69.8
CenterPoint [17]	L	60.3	67.3	59.6	66.8	153.5	80.7
PointPainting [1]	C+L	–	–	65.8*	69.6*	370.0	185.8
PointAugmenting [2]	C+L	66.8 [†]	71.0 [†]	–	–	408.5	234.4
MVP [4]	C+L	66.4	70.5	66.1*	70.0*	371.7	187.1
FusionPainting [50]	C+L	68.1	71.6	66.5	70.7	–	–
AutoAlign [51]	C+L	–	–	66.6	71.1	–	–
FUTR3D [48]	C+L	–	–	64.5	68.3	1069.0	321.4
TransFusion [49]	C+L	68.9	71.6	67.5	71.3	485.8	156.6
BEVFusion (Ours)	C+L	70.2	72.9	68.5	71.4	253.2	119.2

nuScenes

	Frames	mAP/L1	mAPH/L1	mAP/L2	mAPH/L2
AFDetV2-Ens [18]‡	3	84.1	82.6	79.0	77.6
InceptionLiDAR	10	83.8	82.5	79.2	77.8
3DAL-Ens [20]	5	84.6	83.1	79.7	78.2
DeepFusion-Ens [3]‡	5	84.4	83.2	79.5	78.4
MT-Net‡ [55]	3	84.7	83.2	79.9	78.5
MT3D	4	85.0	83.7	80.1	78.7
LIVOX-Detection	7	84.8	83.5	80.2	79.0
MPPNet-Ens‡ [56]	16	85.0	83.7	80.5	79.1
3DAM-Ens	5	85.3	83.8	80.7	79.2
BEVFusion (Ours)†	3	85.7	84.4	80.8	79.5

Waymo

Conclusion

BEVFusion unifies camera and LiDAR features in a shared BEV space that fully preserves both geometric and semantic information.

Achieves superior performance on both nuScenes 3D detection tasks with 1.5- 1.9 $\times$ less computation and 1.3-1.6 $\times$ measured speedup over existing solutions.

BEVFusion also outperforms all existing sensor fusion methods on Waymo open dataset.